

Infinity Hotel Check-In Teacher Notes

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Medium	ID	infinity-hotel-check-in

Big idea: Mathematical concepts can be coherent even when they violate ordinary experience.

Overview

Students act out Hilbert's Hotel to examine why mathematical infinity resists everyday intuition.

Student Prompt

Inspect Infinity Hotel Check-In Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Infinity Hotel Check-In Stimulus A	Student-facing stimulus 1: One new guest arrives at a full infinite hotel; every current guest moves by $n \rightarrow n + 1$. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Infinity Hotel Check-In Stimulus B	Student-facing stimulus 2: Claim card: $\infty + 1 = \infty$ feels impossible until the rule is specified. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Infinity Hotel Check-In Reveal / Twist	Student-facing stimulus 3: Claim card: $\infty + 1 = \infty$ feels impossible until the rule is specified. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Mathematical concepts can be coherent even when they violate ordinary experience.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Infinity Hotel Check-In Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Infinity Hotel Check-In Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Can mathematics know beyond intuition?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how mathematical concepts can be coherent even when they violate ordinary experience.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Use numbered cards on desks or around the room to represent hotel rooms.
- Make clear when students are reasoning about a formal model rather than a physically possible hotel.

Ready-to-Use Examples

- One new guest arrives at a full infinite hotel; every current guest moves by $n \rightarrow n + 1$.
- Claim card: $\infty + 1 = \infty$ feels impossible until the rule is specified.

Suggested Timing

Stage	Teacher move	Watch for
1	Assign students numbered hotel rooms and announce that every room is occupied.	Assumptions, confidence shifts, evidence claims, and language choices.
2	Introduce one new guest and ask whether the hotel can make room.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Use the rule $n \rightarrow n + 1$ so each guest moves to the next room.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Discuss why the model works for countably infinite rooms but not for a finite hotel.	Assumptions, confidence shifts, evidence claims, and language choices.
5	Debrief the difference between everyday imagination and formal definition.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Do not ask whether a real hotel could do this; ask whether the formal rule is coherent.
- Where did your everyday intuition stop helping?

Common Misconceptions

- Students may treat a persuasive pattern as equivalent to proof.
- Students may trust intuition when the formal model uses a different structure.

Assessment Look-Fors

- Distinguishes intuition, empirical pattern, model, and proof.
- Explains why formal reasoning can conflict with first impressions.

Differentiation and Adaptations

- Support: act out the first ten rooms only.
- Challenge: add infinitely many new guests by moving each current guest from room n to room $2n$.

Extension Tasks

- Compare countable infinity with a finite hotel and identify exactly where the rule fails.
- Ask whether mathematical objects need real-world counterparts to produce knowledge.

Related Activities

- Axiom Switch Universe
- Proof versus Pattern
- Certainty Thermometer